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Association of the C-reactive protein-triglyceride glucose index with all-cause and cardiovascular mortality among U.S. adults with metabolic dysfunction-associated steatotic liver disease: a national cohort study

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Abstract

Background: Insulin resistance and systemic inflammation are recognized as primary contributors to the development of metabolic dysfunction-associated steatotic liver disease (MASLD). Nevertheless, the relationship between the C-reactive protein-triglyceride glucose index (CTI), a novel composite marker reflecting both systemic inflammation and insulin resistance, and mortality in MASLD patients has not yet been clearly established.

Methods: Data from 5,029 MASLD patients enrolled in the National Health and Nutrition Examination Survey (1999–2010, 2015–2018) were analyzed. Participants were categorized into CTI quartiles. Associations between CTI, all-cause mortality (ACM), and cardiovascular mortality (CVM) were assessed through multivariate Cox proportional hazards models and restricted cubic spline analyses to evaluate both linear and nonlinear relationships. Mediation and subgroup analyses were conducted to identify factors that mediate or modify the CTI-mortality association. Additional sensitivity analyses were conducted to evaluate the stability and reliability of the findings.

Results: Over a median follow-up of 122 months, a total of 1,072 ACMs and 352 CVMs were observed. Participants in the highest CTI quartile had a significantly elevated risk of ACM (HR: 1.86; 95% CI: 1.50–2.31; $P < 0.001$) and CVM (HR: 2.07; 95% CI: 1.44–2.97; $P < 0.001$) compared to those in the

lowest quartile. Restricted cubic spline analysis revealed a significant nonlinear relationship, with mortality risk rising sharply for CTI values above 8.8. Subgroup analyses revealed significant interactions with age and the fibrosis-4 index, with stronger associations observed in participants younger than 60 years and those with lower fibrosis-4 scores. Exploratory mediation analysis identified several metabolic and inflammatory markers as potential mediators, including neutrophils (mediation proportion: 16%) and glycated hemoglobin (mediation proportion: 33%) for the CTI-ACM link. Adding CTI to a conventional risk model provided the greatest improvement in discriminatory ability for both ACM and CVM compared to CRP or TyG index alone. Sensitivity analyses supported the robustness of the results.

Conclusion: Elevated CTI values are significantly associated with increased ACM and CVM among MASLD patients. CTI may serve as a unique marker for risk stratification within this group.

Keywords: Metabolic dysfunction-associated steatotic liver disease, Cardiovascular disease, NHANES, C-reactive protein, Triglyceride-glucose index, Inflammation, Insulin resistance

Background

Metabolic dysfunction-associated steatotic liver disease (MASLD) is a chronic hepatic condition characterized by excessive lipid accumulation in hepatocytes in individuals without significant alcohol intake or other established causes of liver injury, including viral hepatitis or autoimmune liver disease [1]. In 2023, a multidisciplinary group of specialists recommended the use of the term MASLD to reflect its metabolic basis more clearly [1]. This condition has emerged as the foremost type of hepatic disease globally and is estimated to affect roughly 38% of people worldwide [2]. MASLD is experiencing a growing trend globally, and data suggest that the onset of the disease is shifting to a younger population, allowing more time for the disease to progress towards its complications, including cirrhosis and cardiovascular disease (CVD) [2, 3]. Recent evidence indicates that cardiovascular-related deaths constitute the primary cause of mortality in people affected by this condition [4]. Consequently, identifying MASLD patients at an early stage who are at increased risk of CVD is critically important.

The pathophysiologic processes underlying the link between MASLD and CVD are complex, involving several interrelated factors such as insulin resistance and systemic inflammation [5]. For example, one recent study found that individuals with MASLD who had a high systemic inflammation response index were more likely to experience advanced hepatic fibrosis and cardiovascular mortality (CVM), highlighting the important role that

inflammation plays in the progression of MASLD [6]. Likewise, multiple investigations have demonstrated positive associations between insulin resistance surrogate markers and both advanced hepatic fibrosis and CVM in MASLD, including the triglyceride-glucose (TyG) index [7], the atherogenic index of plasma [8], and the cardiometabolic index [9]. To date, limited research has examined how the combined effects of inflammatory burden and insulin resistance influence outcomes in individuals with MASLD.

The C-reactive protein-triglyceride glucose index (CTI) has emerged over the past few years as a composite metric [10]. It is constructed by integrating C-reactive protein (CRP) with the TyG index, thereby allowing concurrent evaluation of systemic inflammation and insulin resistance [11, 12]. Higher CTI values have been linked to an elevated likelihood of CVD [10], along with increased incidence of heart failure [13] and coronary heart disease [14]. Accumulating evidence indicates that higher CTI levels are progressively associated with an increased likelihood of nonalcoholic fatty liver disease [15]. In addition, rising CTI values have been linked to advancing stages of liver fibrosis, indicating the presence of a clear dose-response association [15]. Compared with CRP or the TyG index considered separately, CTI demonstrates stronger predictive capacity, supporting a synergistic contribution of inflammation and insulin resistance to liver-related damage. Although numerous biomarkers have been investigated, the relationship between CTI and mortality in individuals with MASLD remains

unclear. Accordingly, the present analysis drew on National Health and Nutrition Examination Survey (NHANES) data to examine the associations of CTI with all-cause mortality (ACM) and CVM among U.S. adults diagnosed with MASLD.

Methods

Study Design and Participants

This investigation used information drawn from eight cycles of NHANES, encompassing periods with available CRP measurements from 1999–2010 and 2015–2018. All NHANES-related procedures received ethical clearance from the Review Board of the National Center for Health Statistics prior to data collection. Detailed descriptions of sampling methods, data acquisition, and study protocols are publicly available at <https://www.cdc.gov/nchs/nhanes/index.htm>. Initially, we recruited 81,385 participants. Then, we excluded following participants: (1) participants <18 years old ($n = 34,158$); (2) participants lacking sufficient data to calculate CTI, which includes fasting blood sugar (FBS), CRP, and triglyceride (TG) ($n = 27,131$); (3) participants with incomplete mortality data ($n = 1,637$); (4) participants who did not fulfill the MASLD criteria, including those with incomplete information to compute United States fatty liver index (USFLI), $USFLI < 30$, without cardiometabolic risk factors, or with previous liver disease ($n = 13,078$); and (5) participants with incomplete covariate data (n

= 352). The steps involved in participant inclusion are illustrated in Figure 1.

MASLD definition

According to the Delphi panel criteria, MASLD is diagnosed when hepatic steatosis is present in combination with one or more abnormalities related to cardiometabolic health. The definition excludes people whose liver injury stems from viral, autoimmune, genetic, medication-related, or alcohol-associated causes, with alcohol use thresholds of ≥ 30 g/day for men and ≥ 20 g/day for women. In addition, a USFLI value of 30 or above is considered a reliable indicator of steatotic liver disease [16].

USFLI =

$$\frac{e^{(-0.8073 \times \text{non-Hispanic black} + 0.3458 \times \text{Mexican American} + 0.0093 \times \text{age} + 0.6151 \times \ln(\text{GGT}) + 0.0249 \times \text{waist circumference} + 1.1792 \times \ln(\text{insulin}) + 0.8242 \times \ln(\text{FBS}) - 14.7}}{1 + e^{(-0.8073 \times \text{non-Hispanic black} + 0.3458 \times \text{Mexican American} + 0.0093 \times \text{age} + 0.6151 \times \ln(\text{GGT}) + 0.0249 \times \text{waist circumference} + 1.1792 \times \ln(\text{insulin}) + 0.8242 \times \ln(\text{FBS}) - 14.7}}} \times 100$$

The USFLI has a scale from 0 to 100. For the race-related part, people who identify as “Non-Hispanic Black” or “Mexican American” are given a score of 1, while all other racial groups are coded as 0. Cardiometabolic abnormalities were defined using several criteria [8, 9]. First, excess body fat, either overall or around the waist, was present if body mass index (BMI) was 25 kg/m^2 or higher, or if waist size was more than 94 cm for men and 80 cm for women. Second, problems with glucose regulation were marked by fasting glucose (FBS) of 100 mg/dL or more, glycated hemoglobin (HbA1c) of at least 5.7%, or a diagnosis of diabetes. Third, high blood

pressure was defined as 130/85 mmHg or higher, or if participants reported using anti-hypertensive medication. Fourth, increased TG levels were recorded if concentrations were 150 mg/dL or higher, as well as in individuals receiving anti-hyperlipidemic agents. Finally, low high-density lipoprotein cholesterol (HDL-C) was defined as 50 mg/dL or less in women and 40 mg/dL or less in men.

Calculation of CTI

The CTI value was derived by applying the following computational formula:

$$CTI = 0.412 \times \ln(CRP \text{ [mg/L]}) + \ln(TG \text{ [mg/dl]} \times FBS \text{ [mg/dl]}/2) \text{ [11, 17].}$$

Participants were grouped into four CTI quartile strata (Q1-Q4), using Q1 as the comparison baseline for all analyses.

Definition of covariates

Diabetes was determined according to self-report, current treatment with anti-hyperglycemic medication or insulin, FBS ≥ 126 mg/dL, or HbA1c $\geq 6.5\%$ [18]. Participants were classified as having prediabetes if their FBS fell between 100 and 125 mg/dL or if their HbA1c ranged from 5.7% to 6.4%, as long as they did not report a diabetes diagnosis or the use of glucose-lowering medication [19]. Hypertension was identified when systolic blood pressure (SBP) was ≥ 130 mmHg, diastolic blood pressure (DBP) was ≥ 80 mmHg, or when individuals reported having hypertension or using anti-hypertensive therapy [20, 21]. For greater accuracy, SBP and DBP were averaged over three measurements. CVD status was determined by

participants' reports of a physician-diagnosed condition, including coronary heart disease, angina, heart failure, myocardial infarction, or stroke.

Smoking status was determined using the question, "Have you ever smoked at least 100 cigarettes in your lifetime?" Participants who answered "no" were classified as never smokers. Those who responded "yes" were further categorized based on the question, "Do you currently smoke?" into former smokers and current smokers. The estimated glomerular filtration rate was calculated using the 2021 race and ethnicity free Chronic Kidney Disease Epidemiology Collaboration creatinine equation [22]. The severity of liver fibrosis was evaluated using the FIB-4 (Fibrosis-4) index, calculated as: $\text{FIB-4} = \text{age (years)} \times \text{aspartate aminotransferase (U/L)} \div [\text{platelet count (10}^9\text{/L)} \times \text{alanine aminotransferase}^{0.5} \text{ (U/L)}]$ [23]. This index is a noninvasive marker for advanced liver fibrosis, with values <1.45 indicating low risk, scores of 1.45–2.67 representing an intermediate range, and values >2.67 suggesting a high likelihood of advanced fibrosis or cirrhosis [23].

Mortality outcomes of the study population

Survival outcomes were derived from the Public-Use Linked Mortality Files provided by NHANES, with the most recent revision dated December 31, 2019, and confirmed using probabilistic matching against the National Death Index. Outcomes were categorized in complete accordance with the ICD-10. For each participant, the follow-up period spanned from their initial assessment date through the occurrence of an event—death, cessation of participation, or the fixed study endpoint of December 31, 2019 [10]. ACM

encompassed fatalities from all possible causes, whereas CVM included deaths specifically attributed to cardiovascular-related conditions, including heart disease and cerebrovascular events. The mortality information used here was sourced from the CDC's publicly available linked mortality files (<https://www.cdc.gov/nchs/data-linkage/mortality-public.htm>).

Statistical analysis

We followed CDC guidelines and used sampling strata, primary sampling units, and survey weights to accurately represent the NHANES multistage design. For descriptive analysis, we compared continuous variables across CTI quartiles using survey-adapted linear regression, and reported weighted means and standard errors (SE). We analyzed categorical variables with survey-adjusted chi-square tests and presented them as weighted proportions with SE. To study the relationship between CTI and both ACM and CVM, we used Cox proportional hazards models. We chose covariates based on previous studies [8, 10] and directed acyclic graph reasoning (see Supplemental Figure 1). We constructed three models: Model 1 was unadjusted; Model 2 was adjusted for age, sex, and race/ethnicity; and Model 3 was further adjusted for education level, lifestyle factors (smoking status and alcohol use), BMI, comorbid conditions (hypertension, cancer, and cardiovascular disease), and use of anti-hypertensive and anti-hyperlipidemic medications. We evaluated potential multicollinearity with the variance inflation factor, ensuring that all values remained <5 (Supplemental Tables 1-2). To evaluate the

incremental prognostic value of CTI, CRP, and TyG beyond conventional risk factors for ACM and CVM, we calculated Harrell's C-index. We then used design-based likelihood ratio tests to determine the specific contribution of these biomarkers when added to the baseline model (Model 3). Additionally, to characterize possible nonlinear dose-response patterns between CTI and mortality endpoints, restricted cubic spline (RCS) analyses were carried out using the same covariate adjustments as in Model 3. All RCS models were fitted using survey-weighted Cox regression with four knots placed at the 5th, 35th, 65th, and 95th percentiles of the CTI distribution, using the 35th percentile as the reference. We also used piecewise Cox models to check if the association between CTI and mortality changed above or below the identified inflection point.

We performed subgroup analyses to understand better how CTI relates to mortality in groups based on sex, age, glucose regulation, hypertension, BMI, smoking status, and FIB-4 levels. To explore potential pathways linking CTI with mortality in individuals with MASLD, we conducted exploratory mediation analyses. All models were fitted using survey-weighted Cox proportional hazards regression to account for the complex sampling design of NHANES, including sampling weights, stratification, and clustering. For each candidate mediator, we first estimated the total effect of CTI on mortality using a model without the mediator, and then estimated the direct effect from a model that included the mediator while adjusting for the same covariates. The indirect effect was approximated as the difference

between the total and direct effects, and the proportion mediated was calculated as the indirect effect divided by the total effect. To obtain valid confidence intervals while preserving the survey design, we performed 1,000 stratified bootstrap replicates, resampling within strata and reconstructing the survey design for each replicate. The 2.5th and 97.5th percentiles of the bootstrap distribution were used as the 95% confidence intervals for mediation estimates [24]. Because CTI reflects both inflammation and metabolic dysfunction, which are essential in MASLD progression and mortality [5], we included possible mediators such as markers of glycemic dysregulation (HbA1c, insulin), systemic inflammation (white blood count [WBC], neutrophil and lymphocyte counts, red cell distribution width [RDW]), liver injury or oxidative stress (gamma-glutamyl transferase [GGT], alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase [ALP], total bilirubin, uric acid), and lipid metabolism (HDL-C).

To strengthen our findings, we did five sensitivity analyses. First, the analyses were conducted again after removing individuals who experienced death within the first two years of follow-up. This approach was applied to limit potential bias arising from early mortality, which could have influenced the findings. Second, we adjusted for diabetes and markers of kidney and liver function (estimated glomerular filtration rate and FIB-4). Third, we used Fine-Gray competing-risk models to account for deaths from non-cardiovascular causes. Fourth, to further reduce the potential for reverse

causation, individuals with CRP levels exceeding 10 mg/L were excluded from the analysis. Fifth, to reduce potential misclassification associated with USFLI, we used ultrasonic examination data from the NHANES III cohort (1988–1994). In this cohort, hepatic steatosis was determined using ultrasonic examinations performed with the Toshiba Sonolayer SSA-90A (Toshiba America Medical Systems, Tustin, CA) [25]. Participants with hepatic steatosis, including mild, moderate, and severe cases, were classified as having steatotic liver disease. The remaining participant selection procedures were similar to those used in the original cohort. Finally, 2,669 participants with MASLD from NHANES III were included.

All statistical analyses were done using R software (version 4.5.1; R Foundation for Statistical Computing, Vienna, Austria). Results were considered statistically significant if the two-sided P-value was below 0.05.

Results

Baseline Characteristics Across CTI Quartiles

A total of 5,029 adults aged 18 or older met the study's criteria and were enrolled (see Figure 1). The average participant age was 52.24 years, and 59.42% were male. Table 1 shows baseline characteristics per CTI quartile. People in the highest CTI quartile were more likely to be female, had lower poverty income ratios, and drank less alcohol than those in the lowest quartile. Body composition measures like waist circumference and BMI increased across quartiles. Diabetes was much more common in the top

quartile (50.0%) than the lowest (14.8%), and this group also had higher HbA1c and greater insulin resistance by Homeostatic Model Assessment for Insulin Resistance, showing more severe metabolic issues. Participants with higher CTI values also had higher SBP and DBP, higher total cholesterol, and used anti-hyperlipidemic and anti-hypertensive medications more often, while HDL-C was lower. Markers of systemic inflammation, such as WBC count, neutrophils, lymphocytes, and RDW, increased with higher CTI quartiles. Hepatic biomarkers showed mixed results: GGT and ALP were higher with higher CTI, but total bilirubin was lower in the highest quartile. The FIB-4 score was also generally lower in those with higher CTI values.

Association between CTI and mortality outcomes

Over a median observation period of 122 months (interquartile range, 47 to 170 months), a total of 1,072 ACMs and 352 CVMs were recorded (see Supplemental Table 3 for a breakdown of cause-specific mortality in the cohort). When no covariates were included, a one-unit rise in CTI resulted in markedly greater risks of both ACM (hazard ratio [HR]: 1.25; 95% confidence interval [CI]: 1.16-1.35; $P < 0.001$) and CVM (HR: 1.27; 95% CI: 1.10-1.47; $P: 0.002$). Following adjustment for age, sex, and race, the association between CTI and mortality showed a stronger positive direction. Each unit increment in CTI was linked to a higher risk of ACM (HR: 1.35; 95% CI: 1.24-1.48; $P < 0.001$) and CVM (HR: 1.42; 95% CI: 1.21-1.67; $P < 0.001$). When more demographic and clinical factors were included in Model 3, these links weakened slightly, but CTI still predicted both ACM (HR:

1.30; 95% CI: 1.19–1.41; $P < 0.001$) and CVM (HR: 1.37; 95% CI: 1.17–1.60; $P < 0.001$). People in the highest CTI quartile had much higher risks of ACM (HR: 1.86; 95% CI: 1.50–2.31; $P < 0.001$) and CVM (HR: 2.07; 95% CI: 1.44–2.97; $P < 0.001$) than those in the lowest quartile after full adjustment. Notably, these risk magnitudes for CTI exceeded those observed in the highest quartiles of its individual components. For the highest versus lowest quartile of the TyG index, the risks were lower for both ACM (HR: 1.30; 95% CI: 1.08–1.57; $P = 0.006$) and CVM (HR: 1.55; 95% CI: 1.08–2.22; $P = 0.018$). Similarly, for the highest versus lowest quartile of CRP, the risk was also lower for ACM (HR: 1.72; 95% CI: 1.27–2.32; $P < 0.001$) and was not statistically significant for CVM (HR: 1.32; 95% CI: 0.75–2.33; $P = 0.34$) (Table 2; Supplemental Table 4). All trend analyses for CTI were statistically significant, showing a clear dose-response relationship.

The discriminatory performance of the models was evaluated using Harrell's C-index (Table 3). For ACM, the basic model had a C-index of 0.806. The addition of CRP (0.806; $P = 0.013$) and the TyG index (0.806; $P = 0.001$) resulted in minimal changes in discrimination, whereas the inclusion of CTI yielded the highest C-index (0.808; $P < 0.001$). For CVM, the basic model had a C-index of 0.835. Adding CRP did not significantly change model discrimination (0.835; $P = 0.27$), while the TyG index produced a modest increase (0.839; $P = 0.004$) and CTI yielded the highest C-index (0.840; $P < 0.001$).

Non-linear association between CTI and mortality outcomes

After adjusting for all covariates, RCS analysis showed a strong dose-response link between CTI and both ACM and CVM in people with MASLD (P for overall <0.001). Figure 2 shows that this relationship is clearly non-linear (P for nonlinearity <0.001), with the association becoming stronger at higher CTI values. To describe this further, piecewise Cox models found inflection points at CTI = 8.78 for ACM and CTI = 8.85 for CVM. Below 8.78, the link with ACM was not significant, but above this level, the risk increased sharply (HR per 1-unit: 1.52; 95% CI: 1.34–1.74; P < 0.001). A similar pattern was seen for CVM, with no significant association below CTI = 8.85 and a clear increase above it (HR per 1-unit: 1.65; 95% CI: 1.36–2.01; P < 0.001) (Table 4).

Subgroup analyses

Subgroup analyses were applied to evaluate whether the association between CTI and mortality varied among predefined participant subgroups (see Figures 3-4). For ACM, the positive association remained in almost every category. We found that age (P for interaction <0.001) and FIB-4 score (P for interaction = 0.004) significantly affected this relationship. In adults under 60, ACM increased more sharply with higher CTI (HR per 1-unit: 1.46, 95% CI: 1.22–1.75; P <0.001), while the increase was smaller in those 60 and older (HR per 1-unit: 1.22, 95% CI: 1.10–1.35; P <0.001). The same trend appeared with FIB-4 scores: people with low FIB-4 had the strongest association (HR per 1-unit: 1.48, 95% CI: 1.30–1.67; P <0.001), followed by those with intermediate scores (HR per 1-unit: 1.23, 95% CI:

1.07–1.42; $P = 0.004$). There was no significant link in the high FIB-4 group (HR per 1-unit: 0.88, 95% CI: 0.68–1.13; $P = 0.31$). While we did not find significant interactions with other factors, the association was stronger among patients with abnormal glucose metabolism, hypertension, high BMI, or smokers. For CVM, we saw no significant interaction, but the link was stronger among those with abnormal glucose metabolism, hypertension, low-risk FIB-4 scores, and people who never smoked.

Mediation analyses

We used mediation analyses (see Supplemental Table 5) to investigate how CTI may be linked to ACM and CVM in MASLD patients. For ACM, we found that glycemic indicators HbA1c (33%) and insulin (5%), inflammatory markers WBC (13%), neutrophils (16%), and RDW (8%), as well as liver biomarkers ALP (9%) and GGT (8%), significantly mediated the relationship. For CVM, only inflammatory markers WBC (11%), neutrophils (17%), and RDW (7%), along with liver biomarker ALP (8%), were significant mediators. Glycemic indicators did not show significant mediation for CVM.

Sensitivity analyses

To strengthen confidence in the stability of the findings, further sensitivity evaluations were carried out. As an initial step, participants who experienced death during the first two years of observation were removed from the analytical cohort. We also included the FIB-4 score, estimated glomerular filtration rate and diabetes as additional covariates. In each scenario, CTI maintained a statistically significant relationship with both

ACM and CVM, lending further support to the main findings (see Supplemental Tables 6–7). Moreover, the Fine–Gray competing-risk analysis continued to show a positive and significant link between CTI and CVM after comprehensive adjustment. This pattern was consistent with the estimates obtained from the primary Cox proportional hazards analyses, suggesting that accounting for competing events did not meaningfully alter the association (HR per 1-unit = 1.29, 95% CI = 1.12–1.48, $P < 0.001$) (Supplemental Table 8). Additionally, after excluding participants with CRP levels >10 mg/L, CTI remained significantly associated with both ACM and CVM (Supplemental Table 9).

Finally, a total of 2,669 participants with sonographically defined MASLD from the NHANES III cohort were analyzed (Supplemental Table 10). Over a median follow-up of 314 months (interquartile range, 226 to 336 months), a total of 1,123 ACM and 373 CVM occurred. After full covariate adjustment, each 1-unit increase in CTI was independently associated with both ACM (HR per 1-unit = 1.25, 95% CI = 1.10–1.42, $P < 0.001$) and CVM (HR per 1-unit = 1.37, 95% CI = 1.09–1.72, $P = 0.007$) (Supplemental Table 11).

Discussion

To our knowledge, this study is the first to examine the link between CTI and long-term mortality among U.S. adults with MASLD using a nationally representative sample. Elevated CTI levels corresponded to greater mortality risk, and individuals in the top quartile experienced substantially

higher hazards—approximately 86% for ACM and 107% for CVM— relative to those in the lowest quartile. When visualized with RCS, the CTI-mortality relationship followed a nonlinear pattern. Threshold analyses further supported this shape, revealing turning points at CTI values of 8.78 for ACM and 8.85 for CVM. These associations remained robust across multiple sensitivity analyses, including competing-risk modeling, extra adjustments for estimated glomerular filtration rate and FIB-4, and after removing early deaths. Subgroup analyses showed similar effects overall, but there were significant interactions for ACM by age and FIB-4 score. Exploratory mediation analyses suggested that metabolic derangement and inflammation are key pathways linking CTI to mortality. Importantly, adding CTI to the basic risk model improved its ability to predict risk more than adding CRP or the TyG index alone for both ACM and CVM.

Several inflammatory, metabolic, and lipid-related markers, including CRP, TyG-based measures, and atherogenic index of plasma, are associated with increased CVD risk and mortality within different populations [8, 26-28]. In the MASLD cohort, each additional CTI unit corresponded to a 30% increase in ACM and a 37% increase in CVM. In contrast, Sun et al. [10] reported that, in a general U.S. adult population, each additional CTI unit was associated with nearly a 50% higher risk of ACM and over a 60% higher risk of CVM. The predictive value of CTI for mortality appears lower in MASLD patients, likely due to pre-existing elevated insulin resistance and chronic inflammation resulting from hepatic steatosis [29]. As these patients

already experience significant metabolic and inflammatory stress, the incremental risk conferred by higher CTI is reduced. This pattern was apparent in subgroup analyses stratified by FIB-4 score, in which individuals with higher FIB-4 demonstrated a weaker association between CTI and ACM, while those with low FIB-4 showed a stronger association—suggesting that CTI may serve as an early warning marker for metabolically active disease before advanced fibrosis develops. Our study is consistent with other studies showing increased predictive ability for ACM and CVM when markers of systemic inflammation are combined. Looking at data from 1999-2018 NHANES, it was found that the combined assessment of TyG index and Systemic Inflammation Index was predictive of ACM and CVM, with respective areas under the curve of 0.599 and 0.613 [30]. Notably, combining the TyG index and CRP improves the prediction of cardiovascular events and mortality compared with either marker alone [31], underscoring the advantage of integrating markers that reflect distinct biological pathways. CTI is also associated with increased cancer-related mortality [32], indicating its potential utility in predicting other causes of death. Overall, CTI serves as a clinically relevant marker, but its prognostic value is attenuated in MASLD, warranting judicious assessment in this population.

In this study, adding CTI to the basic risk model improved discriminatory performance more than adding the TyG index or CRP alone for mortality outcomes in patients with MASLD; however, the observed improvements in Harrell's C-index were modest and should not be overinterpreted as

clinically transformative. Rather, these findings suggest incremental gains in risk stratification within an already well-performing model. Inflammation and insulin resistance represent two major pathways through which CTI predicts adverse outcomes [10]. Inflammatory activation accelerates endothelial dysfunction and oxidative stress [33], while insulin resistance promotes metabolic dysregulation and vascular injury [8, 34]. These pathways reinforce each other, creating a metabolic-inflammatory cycle that increases CVD vulnerability [10, 13]. This interaction is particularly prominent in MASLD, where hepatic steatosis drives both systemic insulin resistance and chronic inflammation [8, 26, 35]. In our mediation analysis, HbA1c mediated the largest proportion of the association between CTI and ACM, followed by neutrophil count and WBC count. For CVM, neutrophil count and WBC count were the strongest mediators, highlighting that abnormal metabolism and immune activation are central pathways linking elevated CTI to adverse outcomes. Additional mediation by GGT and ALP suggests that metabolic-inflammatory liver stress further contributes to risk. Taken together, these results support the biological relevance of CTI as an integrative marker of metabolic-inflammatory burden, even when its incremental contribution to model discrimination is limited. These findings align with those reported in previous studies [10, 31] showing that markers integrating metabolic and inflammatory pathways better predict adverse outcomes than single-pathway markers.

The link between higher CTI and increased mortality risk was consistently observed among diverse demographic and clinical subgroups of individuals with MASLD. Adults under 60 had much higher HRs for ACM than those over 60 (P for interaction <0.001). This finding matches earlier studies that found stronger connections between CTI and both mortality [10] and CVD incidence [36] among younger individuals in the general population. Similar results have been seen in people at the early stages of cardiovascular-kidney-metabolic syndrome [37]. Multiple biological and clinical pathways could account for the modifying influence of age on the association between CTI and mortality outcomes. Younger adults often exhibit more active metabolic-inflammatory signaling [38, 39]. Since CTI measures both inflammation and problems with insulin signaling, it may be a better marker of overall metabolic imbalance in younger groups. Also, younger patients usually have a lower baseline risk for CVD, since conditions like hypertension, diabetes, and obesity tend to become more common with age [40], which may make CTI less important in older adults. In addition, permanent changes from years of exposure to cardiovascular risk factors, such as shifts in immune pathways [41], adipose tissue regulation [42], and vascular remodeling [43], may further reduce the predictive value of CTI later in life.

The segmented Cox regression analysis showed that the risk of death linked to CTI in people with MASLD changes sharply at a CTI value of 8.80. Above this value, the risk of mortality increases, while below it, there is no

significant connection. Differences in molecular signaling pathways may explain this nonlinear pattern. Lower CTI values usually mean less inflammation and milder insulin resistance. In these cases, the nuclear factor kappa B pathway is less active, and monocyte adhesion is reduced, which lowers the risk of atherosclerosis [44-47]. In contrast, higher CTI values are linked to ongoing inflammation and insulin resistance, which encourage macrophage polarization and increase new blood vessel growth and fibrosis through interleukin-6 and tumor necrosis factor alpha interactions [48]. These changes mark a shift from blood vessels that can still recover to injuries that lead to lasting structural changes. Inflammation and insulin resistance also strengthen each other, creating a feedback loop that raises cardiovascular risk and marks a point where the body's usual balancing mechanisms start to fail [35, 44-47]. Identifying this CTI threshold can help spot high-risk patients early and show the importance of timely prevention, such as lifestyle changes, for MASLD patients with CTI values above this level.

This study has several limitations. First, a considerable portion of participants was removed because key variables, including USFLI, were incomplete. The decrease in the number of included participants could have led to selection bias and reduced the generalizability of the findings to wider populations. Second, MASLD diagnosis was based on the USFLI index rather than imaging or liver biopsy, which may contribute to misclassification bias. Third, despite comprehensive adjustment for

confounding variables, unmeasured confounders may persist. Fourth, acute or chronic inflammatory states, as indicated by transiently elevated CRP, were not considered in the baseline survey. Fifth, the analysis focused solely on individuals from the United States, making it uncertain whether the findings apply to other demographic or geographic groups. Sixth, CVM data rely on death certificate coding, which may contribute to misclassification bias. Seventh, the potential influence of pharmacologic therapies on CTI was not fully evaluated. During the study period, several new medication classes targeting glucose metabolism, lipid profiles, and systemic inflammation became widely used, and these agents may have modified the CTI-mortality link. Eighth, cirrhosis was not evaluated as a competing risk for ACM and CVM because this outcome was not available, which may have biased risk estimates if cirrhosis-related mortality occurred before the events of interest. Finally, only baseline CTI values were evaluated, without monitoring dynamic fluctuations in CRP, TG, and FBS. Future multicenter, long-term, longitudinal studies involving large, heterogeneous cohorts are necessary to determine whether CTI can serve as a reliable screening metric or risk marker in populations with MASLD.

Conclusion

This investigation shows that higher CTI values in people diagnosed with MASLD correspond to a greater likelihood of ACM and CVM, particularly among younger patients and those with a low FIB-4 score. A significant nonlinear relationship was identified, with a CTI value of approximately 8.8

representing a critical threshold above which risk increases sharply. Given that CTI is cost-effective and straightforward to measure, it has potential utility as an early screening indicator for identifying patients with MASLD at elevated risk. Additional research is needed to confirm and extend these observations.

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Abbreviations: ACM: all-cause mortality; ALP: alkaline phosphatase; BMI: body mass index; CI: confidence interval; CRP: C-reactive protein; CTI: C-reactive protein-triglyceride glucose index; CVD: cardiovascular disease; CVM: cardiovascular mortality; DBP: diastolic blood pressure; FBS: fasting

blood sugar; FIB-4: Fibrosis-4 index; GGT: gamma-glutamyl transferase; HbA1c: glycated hemoglobin; HDL-C: high-density lipoprotein cholesterol; HR: hazard ratio; MASLD: metabolic dysfunction-associated steatotic liver disease; NHANES: National Health and Nutrition Examination Survey; RCS: restricted cubic spline; RDW: red cell distribution width; SBP: systolic blood pressure; SE: standard error; TG: triglyceride; TyG: triglyceride-glucose; USFLI: United States fatty liver index; WBC: white blood count.

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Data availability

The current study used data from the National Health and Nutrition Examination Survey (NHANES) database. For comprehensive details regarding the database and its methods, readers are invited to check the official NHANES website at: <http://www.cdc.gov/nchs/nhanes/>.

Ethical approval and consent to participate:

The research protocol for NHANES was officially approved by the

Institutional Review Board of the National Center for Health Statistics (NCHS IRB). All participants provided informed written consent. As this study utilizes secondary analysis of deidentified, publicly accessible data from NHANES, it was exempt from additional review by the institutional review board according to the standards for secondary data analysis.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

Authors' contributions

RA and KH were responsible for conception and design. RA, FJ, MJ, NS, and SAS contributed to data extraction, proofreading, and statistical analysis. KN, IA, AFH, TK, TBS, AVH, and JRN were involved in data interpretation and figure mapping. RA, FJ, MJ, NS, and SAS participated in writing the original draft. KN, IA, AFH, TK, TBS, AVH, JRN, and KH critically revised the manuscript. KH conceived and supervised the work. All authors participated in editing, reviewing, and approving the final manuscript.

Figure legends:

Figure 1: Flow diagram of participant selection.

CRP: C-reactive protein; FBS: fasting blood sugar; MASLD: metabolic dysfunction-associated steatotic liver disease; NHANES: National Health and Nutrition Examination Survey; TG: triglyceride; USFLI: United States Fatty Liver Index.

Figure 2: Dose-response curves of C-reactive protein-triglyceride glucose index and mortality outcomes using restricted cubic splines with four knots. Adjusted for age, gender, race, education, smoking, drinking, BMI, hypertension, cancer, CVD, anti-hypertensive agents, and anti-hyperlipidemia agents. BMI: body mass index; CI: confidence interval; CTI: C-reactive protein-triglyceride glucose index; CVD: cardiovascular disease; HR: hazard ratio.

Figure 3: Subgroup analysis of the association between C-reactive protein-triglyceride glucose index and all-cause mortality.

BMI: body mass index; CI: confidence interval; FIB-4: fibrosis-4 index; HR: hazard ratio.

Figure 4: Subgroup analysis of the association between C-reactive protein-triglyceride glucose index and cardiovascular mortality.

BMI: body mass index; CI: confidence interval; FIB-4: fibrosis-4 index; HR: hazard ratio.

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Table 1. The baseline characteristics stratified by C-reactive protein-triglyceride glucose index quartiles, weighted for representativeness

		CTI				P value
	Total N = 5029	Q1 (<8.32) N = 1258	Q2 (8.32 to 8.87) N = 1257	Q3 (8.87 to 9.46) N = 1257	Q4 (>9.46) N = 1257	
Age, years	52.24 (0.34)	52.06 (0.63)	51.16 (0.51)	53.37 (0.53)	52.38 (0.60)	0.20
Male, %	59.42 (0.94)	70.19 (1.79)	62.33 (1.78)	54.83 (2.08)	50.23 (1.76)	<0.001
Race, %						0.27
Mexican American	12.10 (1.07)	12.46 (1.34)	11.66 (1.21)	11.80 (1.23)	12.47 (1.57)	
Other Hispanic	5.41 (0.63)	4.29 (0.68)	5.06 (0.94)	5.04 (0.82)	7.26 (0.96)	
Non-Hispanic White	71.26 (1.46)	72.28 (2.00)	72.54 (1.91)	71.51 (1.92)	68.70 (2.18)	
Non-Hispanic Black	6.26 (0.51)	6.41 (0.76)	6.31 (0.65)	5.76 (0.63)	6.54 (0.86)	
Other Race	4.98 (0.42)	4.56 (0.78)	4.44 (0.74)	5.90 (0.91)	5.04 (0.81)	
Marital status, %						0.22
Not married, widowed, divorced, or separated	30.86 (1.01)	28.86 (1.78)	29.60 (1.67)	31.09 (1.71)	33.90 (2.33)	
Married or living with a partner	69.14 (1.01)	71.14 (1.78)	70.40 (1.67)	68.91 (1.71)	66.10 (2.33)	
Education level, %						0.11
≤High school	48.22 (1.24)	44.77 (2.17)	47.36 (1.96)	51.53 (2.12)	49.23 (2.27)	
>High school	51.78 (1.24)	55.22 (2.17)	52.64 (1.96)	58.47 (2.12)	50.77 (2.27)	
Poverty income ratio	3.01 (0.043)	3.18 (0.062)	3.12 (0.064)	2.90 (0.073)	2.82 (0.078)	<0.001
Smoking, %						0.23
Never	47.96 (1.07)	50.99 (2.02)	48.25 (1.87)	47.20 (2.11)	45.37 (1.82)	
Former	33.76 (0.98)	32.49 (1.82)	34.69 (1.72)	32.32 (2.18)	35.54 (1.84)	
Current	18.29 (0.73)	16.53 (1.27)	17.06 (1.35)	20.48 (1.49)	19.10 (1.24)	
Drinking, %						<0.001
Yes	68.82 (1.11)	76.07 (1.60)	70.48 (1.84)	67.50 (2.09)	61.16 (2.08)	
No	31.18 (1.11)	23.93 (1.60)	29.52 (1.84)	32.50 (2.09)	38.84 (2.08)	
BMI, kg/m ²	33.87 (0.14)	31.80 (0.20)	33.87 (0.26)	34.63 (0.25)	35.19 (0.32)	<0.001
Waist circumference, cm	112.84 (0.32)	108.57 (0.47)	112.72 (0.58)	114.42 (0.51)	115.69 (0.70)	<0.001
Hypertension, %	70.39 (1.07)	68.64 (1.80)	67.81 (1.95)	72.05 (2.02)	73.11 (1.78)	0.11
Glucose metabolism status, %						<0.001
Normal	19.82 (0.89)	30.06 (1.82)	25.29 (1.74)	14.47 (1.39)	9.35 (1.20)	
Prediabetes	51.40 (1.01)	55.11 (1.79)	52.48 (1.80)	57.31 (1.91)	40.62 (2.04)	
Diabetes	28.78 (0.79)	14.82 (1.26)	22.23 (1.48)	28.22 (1.59)	50.03 (1.92)	
Anti-hyperlipidemic agents, %	26.43 (0.79)	23.72 (1.60)	25.31 (1.54)	27.03 (1.56)	29.69 (1.56)	0.049
Anti-hypertensive agents, %	41.76 (1.00)	39.22 (1.74)	38.75 (1.80)	44.52 (2.10)	44.60 (2.01)	0.036
SBP, mm Hg	126.37 (0.31)	125.17 (0.55)	125.31 (0.55)	126.81 (0.72)	128.21 (0.60)	<0.001
DBP, mm Hg	72.59 (0.32)	71.57 (0.55)	73.04 (0.47)	72.05 (0.50)	73.72 (0.58)	0.014
eGFR, ml/min/1.73m ²	94.14 (0.50)	94.62 (0.73)	93.93 (0.72)	94.02 (0.79)	93.99 (0.86)	0.56
Uric acid, mg/dL	6.14 (0.027)	6.04 (0.051)	6.21 (0.046)	6.15 (0.051)	6.17 (0.050)	0.12

HbA1c, %	5.95 (0.018)	5.56 (0.018)	5.71 (0.026)	5.90 (0.027)	6.65 (0.055)	<0.00 1
TC, mg/dL	199.05 (1.08)	189.17 (1.42)	195.94 (1.64)	201.41 (1.68)	209.78 (2.06)	<0.00 1
HDL-C, mg/dL	46.23 (0.27)	50.84 (0.55)	47.11 (0.48)	45.01 (0.39)	41.93 (0.43)	<0.00 1
LDL-C, mg/dL	118.39 (0.76)	116.31 (1.32)	119.05 (1.45)	121.71 (1.58)	116.13 (1.40)	0.54
HOMA-IR	7.09 (0.11)	5.23 (0.13)	6.37 (0.29)	6.74 (0.21)	10.05 (0.39)	<0.00 1
WBC, × 10 ⁹ / L	7.38 (0.044)	6.73 (0.063)	7.19 (0.075)	7.66 (0.077)	7.93 (0.088)	<0.00 1
Neutrophil, × 10 ⁹ / L	4.41 (0.035)	3.96 (0.047)	4.30 (0.064)	4.58 (0.060)	4.82 (0.067)	<0.00 1
Lymphocyte, × 10 ⁹ / L	2.11 (0.018)	1.96 (0.027)	2.05 (0.023)	2.20 (0.043)	2.22 (0.032)	<0.00 1
RDW, %	13.14 (0.030)	12.81 (0.037)	13.01 (0.035)	13.24 (0.048)	13.49 (0.057)	<0.00 1
GGT, U/L	44.13 (1.11)	42.13 (1.75)	39.12 (1.31)	43.42 (2.86)	51.94 (2.32)	<0.00 1
ALT, U/L	32.66 (0.41)	33.88 (0.78)	32.73 (0.73)	31.86 (0.89)	30.77 (0.76)	0.099
AST, U/L	27.57 (0.32)	29.03 (0.59)	27.17 (0.50)	26.83 (0.67)	27.25 (0.63)	0.052
ALP, U/L	76.40 (0.55)	72.05 (0.88)	73.37 (0.99)	76.93 (0.95)	83.33 (1.45)	<0.00 1
Total bilirubin, mg/dL	0.68 (0.0069)	0.78 (0.014)	0.72 (0.011)	0.64 (0.010)	0.58 (0.013)	<0.00 1
USFLI	53.02 (0.35)	46.31 (0.59)	51.21 (0.74)	53.96 (0.59)	60.66 (0.75)	<0.00 1
FIB-4 score	1.13 (0.016)	1.24 (0.032)	1.10 (0.024)	1.10 (0.026)	1.10 (0.028)	<0.00 1
All-cause mortality, N	1072	291	277	261	243	0.83
Cardiovascular mortality, N	352	93	89	79	91	0.81

HOMA-IR index = Fasting blood sugar (mmol/L) × Fasting insulin (μU/mL)/22.5

ALP: alkaline phosphatase; ALT: alanine aminotransferase; AST: aspartate aminotransferase; BMI: body mass index; CTI: C-reactive protein-triglyceride glucose index; DBP: diastolic blood pressure; eGFR: estimated glomerular filtration rate; FIB-4: fibrosis-4 index; GGT: gamma-glutamyl transferase; HbA1c: glycated hemoglobin; HDL-C: high-density lipoprotein cholesterol; HOMA-IR: homeostasis model assessment of insulin resistance; LDL-C: low-density lipoprotein cholesterol; PIR: poverty income ratio; Q1: first quartile; Q2: second quartile; Q3: third quartile; Q4: fourth quartile; RDW: red cell distribution width; SBP: systolic blood pressure; TC: total cholesterol; USFLI: United States Fatty Liver Index; WBC: white blood cell count

Table 2. Association between C-reactive protein-triglyceride glucose index and mortality

	Model 1 HR (95% CI)	P-value	Model 2 HR (95% CI)	P-value	Model 3 HR (95% CI)	P-value
All-cause mortality						
CTI (per 1-unit)	1.25 (1.16-1.35)	<0.001	1.35 (1.24-1.48)	<0.001	1.30 (1.19-1.41)	<0.001
Q1	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Q2	1.03 (0.83-1.27)	0.81	1.15 (0.93-1.42)	0.19	1.08 (0.88-1.33)	0.44
Q3	1.24 (1.00-1.53)	0.046	1.34 (1.07-1.68)	0.011	1.22 (0.98-1.51)	0.081
Q4	1.78 (1.46-2.17)	<0.001	2.06 (1.64-2.57)	<0.001	1.86 (1.50-2.31)	<0.001
P for trend	<0.001		<0.001		<0.001	
Cardiovascular mortality						
CTI (per 1- unit)	1.27 (1.10-1.47)	0.002	1.42 (1.21-1.67)	<0.001	1.37 (1.17-1.60)	<0.001
Q1	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Q2	1.01 (0.73-1.41)	0.94	1.17 (0.81-1.68)	0.41	1.12 (0.77-1.62)	0.54
Q3	1.15 (0.80-1.64)	0.46	1.31 (0.88-1.94)	0.18	1.22 (0.80-1.87)	0.35
Q4	1.85 (1.33-2.59)	<0.001	2.26 (1.56-3.25)	<0.001	2.07 (1.44-2.97)	<0.001
P for trend	0.001		<0.001		<0.001	

Model 1: unadjusted.

Model 2: Adjusted for age, gender, race.

Model 3: Adjusted for age, gender, race, education, smoking, drinking, BMI, hypertension, cancer, CVD, anti-hypertensive agents, anti-hyperlipidemic agents.

BMI: body mass index; CI: confidence interval; CTI: C-reactive protein-triglyceride glucose index; CVD: cardiovascular disease; HR: hazard ratio; Q1: first quartile; Q2: second quartile; Q3: third quartile; Q4: fourth quartile; Ref: reference

Table 3. Incremental discriminatory performance of C-reactive protein-triglyceride glucose index, triglyceride glucose index, and C-reactive protein added to the basic model for all-cause and cardiovascular mortality

	Harrell's C-index (95% CI)	p-value
All-cause mortality		
Basic model	0.806 (0.793-0.818)	Ref.
Basic model + CRP	0.806 (0.793-0.819)	0.013
Basic model + TyG	0.806 (0.793-0.819)	0.001
Basic model + CTI	0.808 (0.796-0.821)	<0.001
Cardiovascular mortality		
Basic model	0.835 (0.814-0.857)	Ref.
Basic model + CRP	0.835 (0.814-0.857)	0.27
Basic model + TyG	0.839 (0.818-0.860)	0.004
Basic model + CTI	0.840 (0.819-0.861)	<0.001

The basic model incorporates age, gender, race, education, smoking, drinking, BMI, hypertension, cancer, CVD, anti-hypertensive agents, and anti-hyperlipidemic agents.

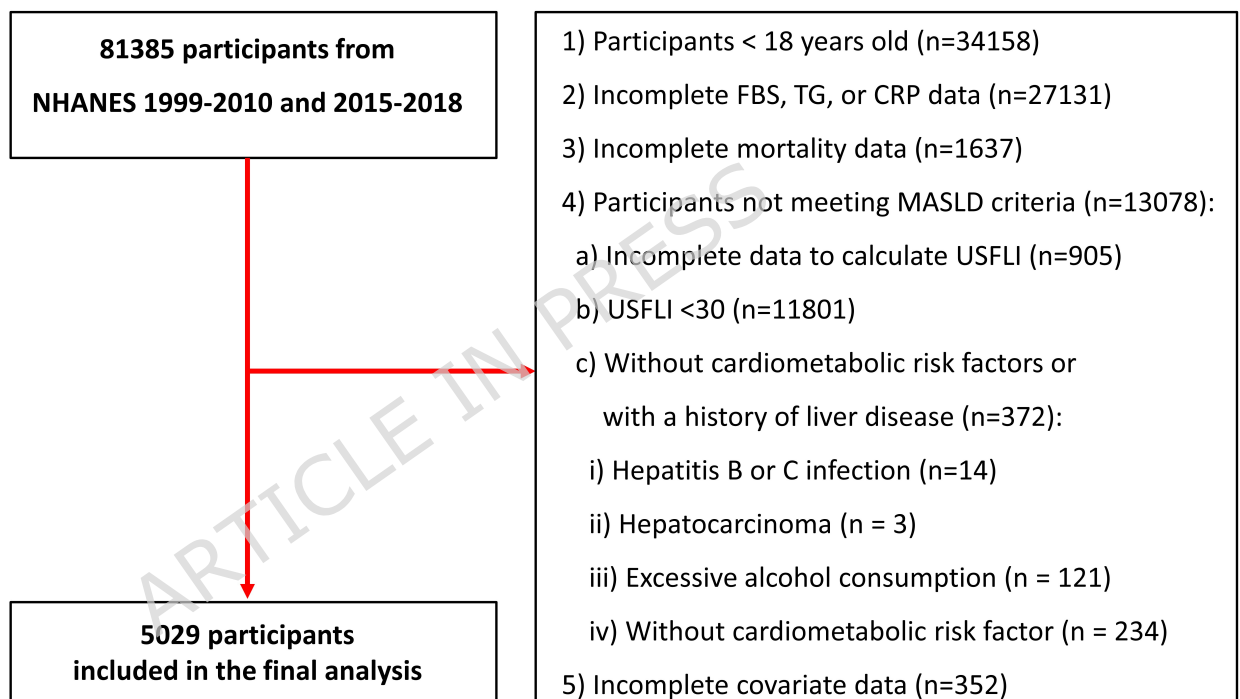
BMI: body mass index; CI: confidence interval; CRP: C-reactive protein; CTI: C-reactive protein-triglyceride glucose index; CVD: cardiovascular disease; Ref: reference; TyG: triglyceride glucose index.

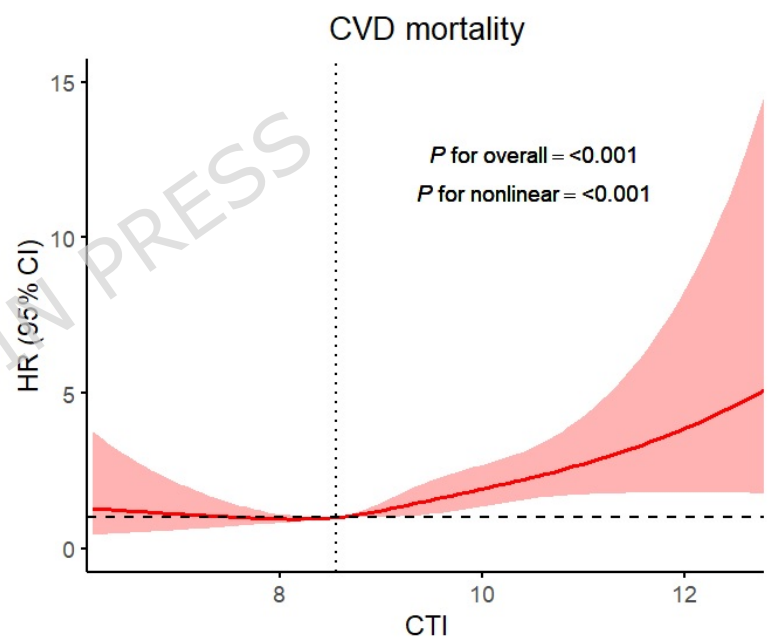
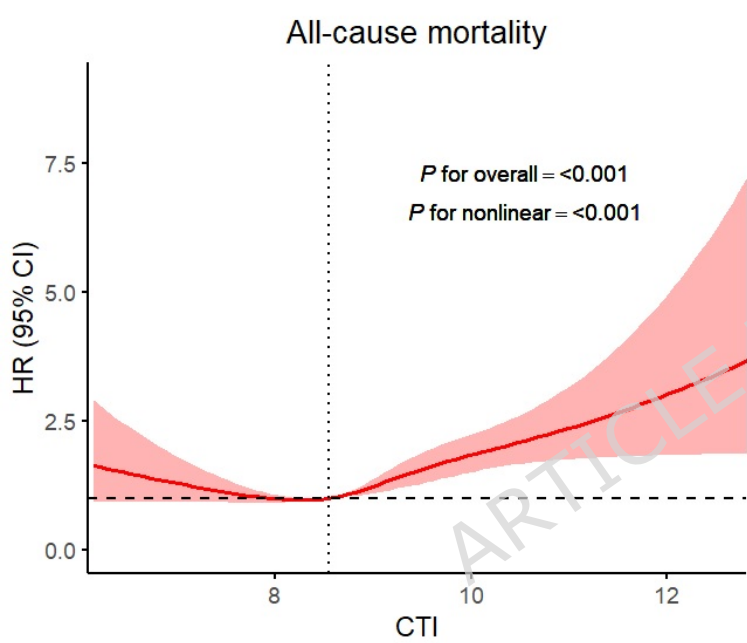
Table 4. C-reactive protein-triglyceride glucose index threshold effects on all-cause and cardiovascular mortality in MASLD

	HR (95% CI)	P- value
All-cause mortality		
Total	1.30 (1.19-1.41)	<0.001
Fitting by two-piecewise Cox proportional risk model		
Inflection point	8.78	
CTI ≤ 8.78	1.02 (0.86-1.21)	0.85
CTI > 8.78	1.52 (1.34-1.74)	<0.001
CVD mortality		
Total	1.37 (1.17-1.60)	<0.001
Fitting by two-piecewise Cox proportional risk model		
Inflection point	8.85	
CTI ≤ 8.85	1.04 (0.77-1.40)	0.82
CTI > 8.85	1.65 (1.36-2.01)	<0.001

Adjusted for age, gender, race, education, smoking, drinking, BMI, hypertension, cancer, CVD, anti-hypertensive agents, anti-hyperlipidemic agents.

BMI: body mass index; CI: confidence interval; CTI: C-reactive protein-triglyceride glucose index; CVD: cardiovascular disease; HR: hazard ratio; MASLD: metabolic dysfunction-associated steatotic liver disease





Variables	n (%)	HR (95 CI%)	P-value	P for interaction
Gender				0.86
Male	2881 (57.3%)	1.29 (1.17–1.43)	<0.001	
Female	2148 (42.7%)	1.31 (1.14–1.50)	<0.001	
Age				<0.001
<60	2755 (54.8%)	1.46 (1.22–1.75)	<0.001	
≥60	2274 (45.2%)	1.22 (1.10–1.35)	<0.001	
Glucose metabolism status				0.65
Normal	863 (17.2%)	1.10 (0.73–1.67)	0.65	
Pre-diabetes	2418 (48.1%)	1.22 (1.02–1.46)	0.026	
Diabetes	1748 (34.7%)	1.19 (1.08–1.32)	<0.001	
Hypertension				0.26
Yes	3599 (71.6%)	1.34 (1.22–1.48)	<0.001	
No	1430 (28.4%)	1.07 (0.78–1.47)	0.65	
BMI				0.12
<25	263 (5.2%)	1.16 (0.75–1.79)	0.5	
≥25	4766 (94.8%)	1.30 (1.18–1.43)	<0.001	
Smoking status				0.74
Never	2430 (48.3%)	1.25 (1.06–1.47)	0.007	
Former	1668 (33.2%)	1.30 (1.10–1.55)	0.003	
Current	931 (18.5%)	1.43 (1.18–1.72)	<0.001	
FIB-4				0.004
Low risk	3571 (71.0%)	1.48 (1.30–1.67)	<0.001	
Intermediate risk	1232 (24.5%)	1.23 (1.07–1.42)	0.004	
High risk	226 (4.5%)	0.88 (0.68–1.13)	0.31	

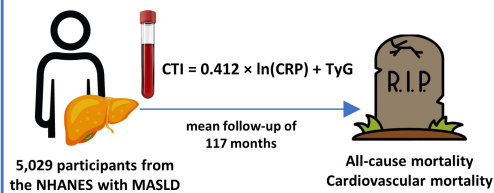


Variables	n (%)	HR (95 CI%)	P-value	P for interaction
Gender				0.43
Male	2881 (57.3%)	1.39 (1.14–1.69)	0.001	
Female	2148 (42.7%)	1.30 (1.02–1.66)	0.033	
Age				0.35
<60	2755 (54.8%)	1.38 (1.00–1.90)	0.051	
≥60	2274 (45.2%)	1.34 (1.10–1.62)	0.003	
Glucose metabolism status				0.16
Normal	863 (17.2%)	0.86 (0.46–1.62)	0.65	
Pre-diabetes	2418 (48.1%)	1.49 (1.09–2.03)	0.013	
Diabetes	1748 (34.7%)	1.33 (1.10–1.62)	0.003	
Hypertension				0.79
Yes	3599 (71.6%)	1.37 (1.15–1.64)	<0.001	
No	1430 (28.4%)	1.46 (0.84–2.53)	0.18	
BMI				0.67
<25	263 (5.2%)	1.92 (1.02–3.61)	0.043	
≥25	4766 (94.8%)	1.35 (1.15–1.59)	<0.001	
Smoking status				0.57
Never	2430 (48.3%)	1.50 (1.23–1.83)	<0.001	
Former	1668 (33.2%)	1.41 (1.06–1.87)	0.017	
Current	931 (18.5%)	1.20 (0.82–1.74)	0.36	
FIB-4				0.18
Low risk	3571 (71.0%)	1.54 (1.24–1.91)	<0.001	
Intermediate risk	1232 (24.5%)	1.18 (0.91–1.53)	0.21	
High risk	226 (4.5%)	1.58 (0.95–2.63)	0.081	

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Association of the C-reactive protein–triglyceride glucose index with all-cause and cardiovascular mortality among U.S. adults with metabolic dysfunction-associated steatotic liver disease: a national cohort study

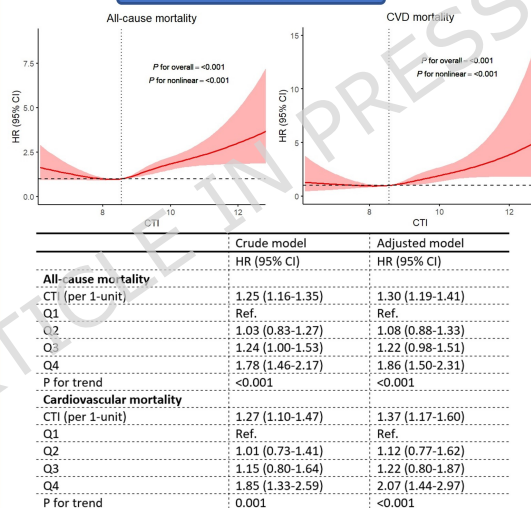
Study design



Statistical analysis

- ✓ Multivariable-adjusted Cox regression analysis
- ✓ Restricted cubic splines
- ✓ Two-piecewise Cox proportional hazards models
- ✓ Harrell's C-index
- ✓ Subgroup and interaction analyses
- ✓ Mediations analyses
- ✓ Sensitivity analyses

Results



Key findings

- ✓ Higher CTI was independently associated with increased all-cause and cardiovascular mortality.
- ✓ A nonlinear dose–response relationship was observed, with mortality risk rising sharply when CTI exceeded 8.8.
- ✓ Adding CTI to the basic risk model improved discrimination for mortality more than CRP or TyG alone.
- ✓ Associations were stronger in younger individuals and those with lower FIB-4 scores.
- ✓ Sensitivity analyses confirmed the robustness of the findings.